

Case study 3:-

Student Performance Analyzer:-A university wants to generate semester reports. Accept marks of five subjects and calculate Total,Average, and Percentage.

Algorithm – Student Performance Analyzer

1. Start.
2. Declare variables: m1, m2, m3, m4, m5, total, average, percentage.
3. Input marks of five subjects.
4. Calculate Total:
 - $\text{total} = m1 + m2 + m3 + m4 + m5$
5. Calculate Average:
 - $\text{average} = \text{total} / 5$
6. Calculate Percentage:
 - $\text{percentage} = (\text{total} / 500) \times 100$
7. Display Total, Average, and Percentage.
8. Stop.

Pseudocode:-

BEGIN

DECLARE m1, m2, m3, m4, m5 AS INTEGER

DECLARE total, average, percentage AS FLOAT

INPUT m1, m2, m3, m4, m5

total = m1 + m2 + m3 + m4 + m5

average = total / 5

percentage = (total / 500) * 100

PRINT "Total = ", total

PRINT "Average = ", average

PRINT "Percentage = ", percentage

END

Test cases:- input/output combination.

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_3.c -o CASE_study_3 ; if ($?) { .\CASE_study_3 }  
enter the five subject marks: 78 85 91 80 76
```

```
Total = 500  
Average = 82.000000  
Percentage = 82.000000%
```

```
PS C:\Users\khush\Desktop\PROJECT 1> █
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_3.c -o CASE_study_3 ; if ($?) { .\CASE_study_3 }  
enter the five subject marks: 79 86 93 82 77
```

```
Total = 500  
Average = 83.000000  
Percentage = 83.000000%
```

```
PS C:\Users\khush\Desktop\PROJECT 1> █
```

```
PS C:\Users\khush\Desktop\PROJECT 1> gcc CASE_study_3.c -o CASE_study_3 ; if ($?) { .\CASE_study_3 }  
enter the five subject marks:  
54 67 78 93 88
```

```
Total = 500  
Average = 76.000000  
Percentage = 76.000000%
```

```
PS C:\Users\khush\Desktop\PROJECT 1> █
```